

Practical 10

```
In [1]: import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
In [2]: data = pd.read_csv("Iris.csv")
```

```
In [3]: data
```

Out[3]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

```
In [4]: data.head()
```

Out[4]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [5]: data.sample(10)
```

Out [5]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
12	13	4.8	3.0	1.4	0.1	Iris-setosa
21	22	5.1	3.7	1.5	0.4	Iris-setosa
64	65	5.6	2.9	3.6	1.3	Iris-versicolor
125	126	7.2	3.2	6.0	1.8	Iris-virginica
136	137	6.3	3.4	5.6	2.4	Iris-virginica
32	33	5.2	4.1	1.5	0.1	Iris-setosa
89	90	5.5	2.5	4.0	1.3	Iris-versicolor
84	85	5.4	3.0	4.5	1.5	Iris-versicolor
62	63	6.0	2.2	4.0	1.0	Iris-versicolor
92	93	5.8	2.6	4.0	1.2	Iris-versicolor

In [6]: `data.shape`

Out [6]: (150, 6)

In [7]: `data.info()`

```
<class 'pandas.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Id               150 non-null   int64
1   SepalLengthCm    150 non-null   float64
2   SepalWidthCm     150 non-null   float64
3   PetalLengthCm    150 non-null   float64
4   PetalWidthCm     150 non-null   float64
5   Species          150 non-null   str
dtypes: float64(4), int64(1), str(1)
memory usage: 7.2 KB
```

In [9]: `data["Species"].value_counts()`

Out [9]:

Species	
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

Name: count, dtype: int64

In [11]:

```
sum_data = data["SepalLengthCm"].sum()
mean_data = data["SepalLengthCm"].mean()
median_data = data["SepalLengthCm"].median()

print("Sum:", sum_data)
print("Mean:", mean_data)
print("Median:", median_data)
```

Sum: 876.5
Mean: 5.8433333333333334
Median: 5.8

In [12]: `data.isnull()`

Out [12]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
	0	False	False	False	False	False
	1	False	False	False	False	False
	2	False	False	False	False	False
	3	False	False	False	False	False
	4	False	False	False	False	False
	...	...	...	...	...	...
	145	False	False	False	False	False
	146	False	False	False	False	False
	147	False	False	False	False	False
	148	False	False	False	False	False
	149	False	False	False	False	False

150 rows × 6 columns

In [13]:

data.isnull().sum()

Out[13]:

Id0

SepalLengthCm0

SepalWidthCm0

PetalLengthCm0

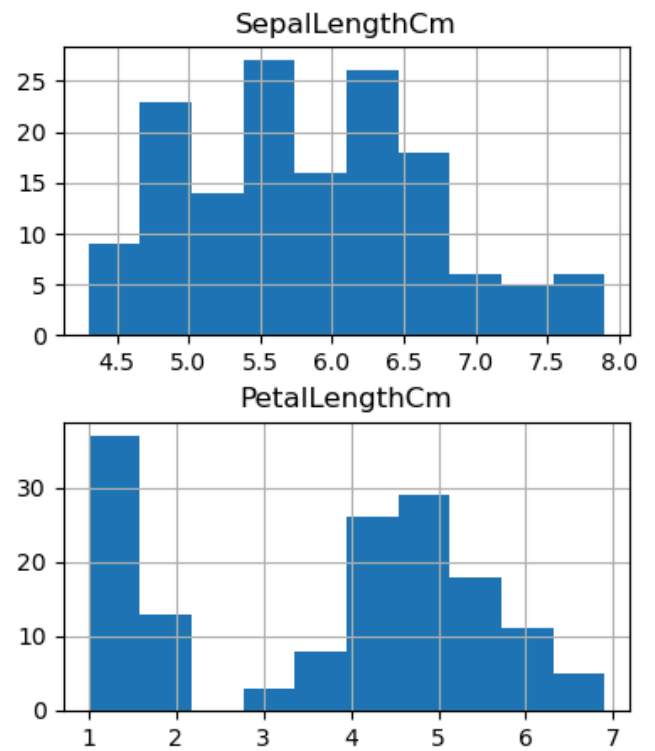
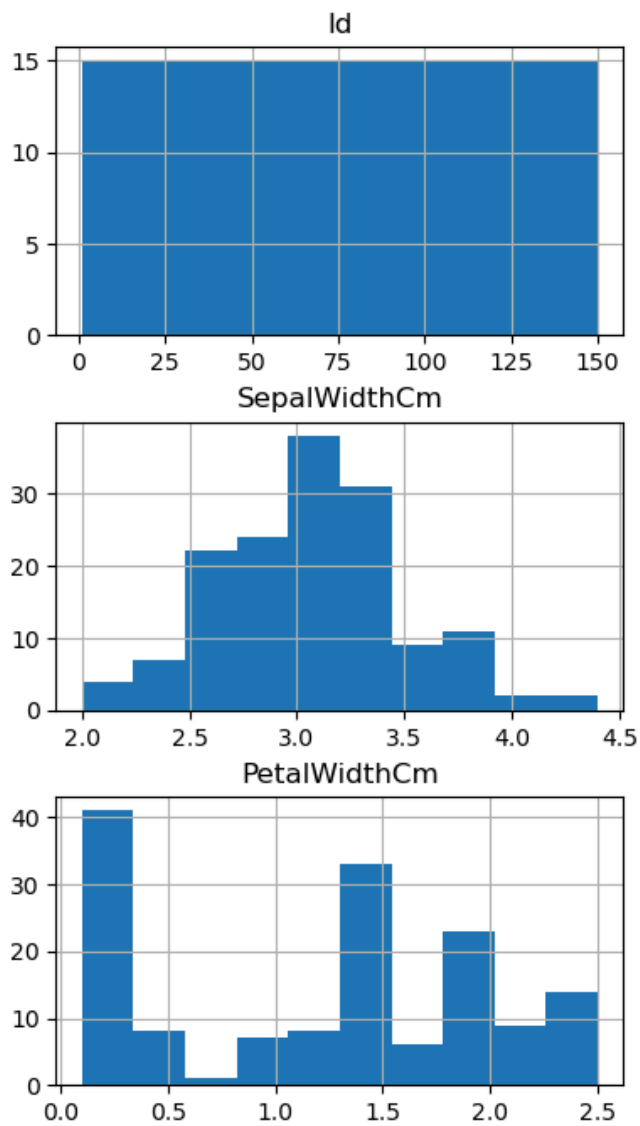
PetalWidthCm0

Species0

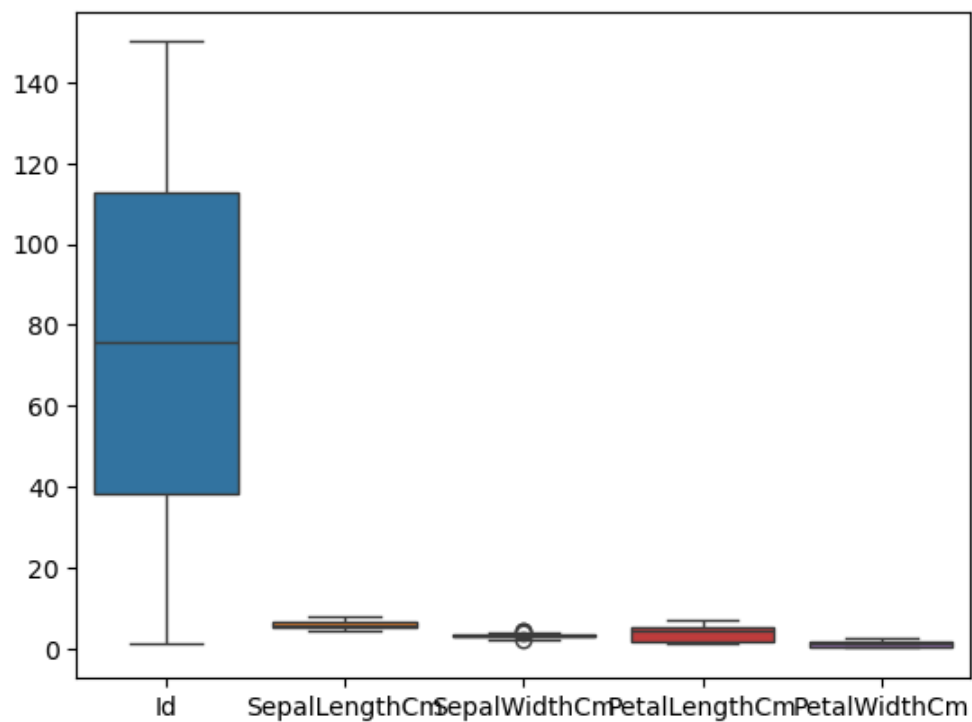
dtype: int64

In [14]:

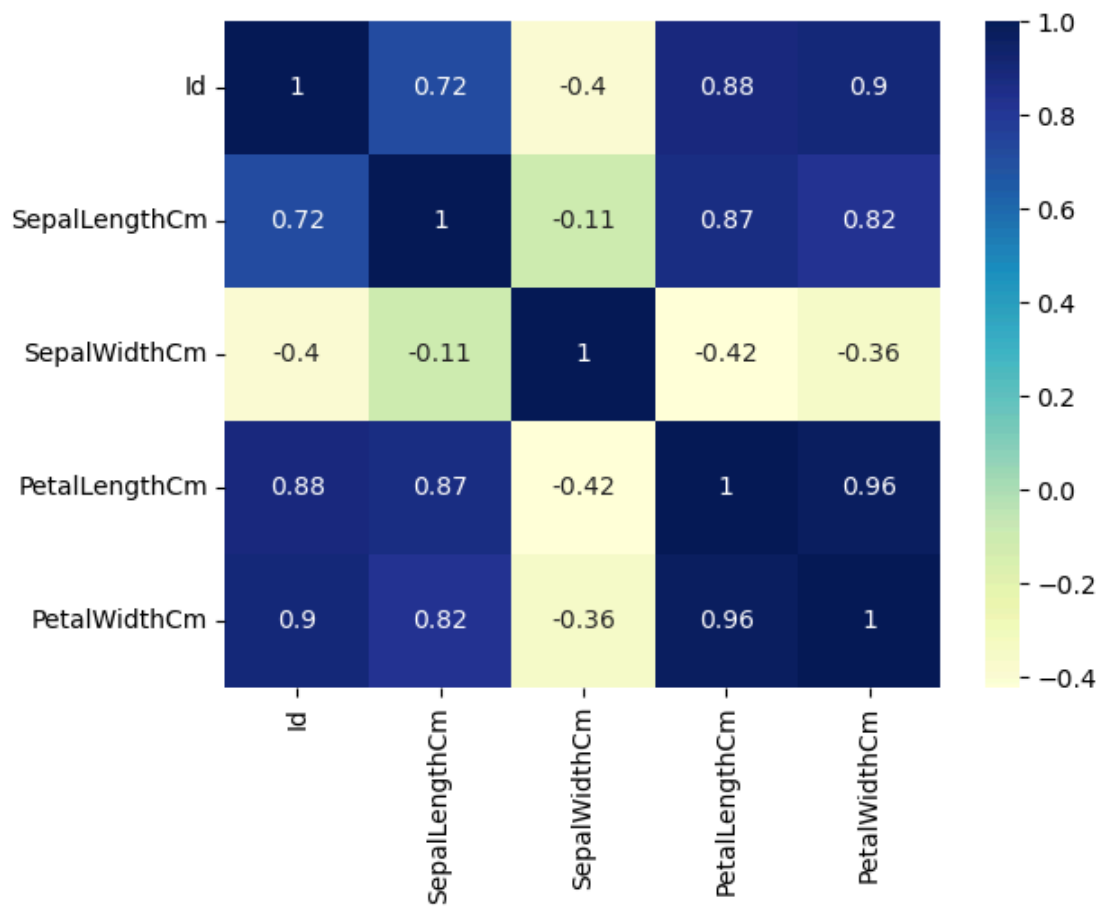
data.hist(figsize=(10,8))
plt.show()



```
In [15]: sns.boxplot(data=data)
plt.show()
```



```
In [17]: sns.heatmap(data.select_dtypes(include=np.number).corr(), cmap="YlGnBu", annot=True)
plt.show()
```

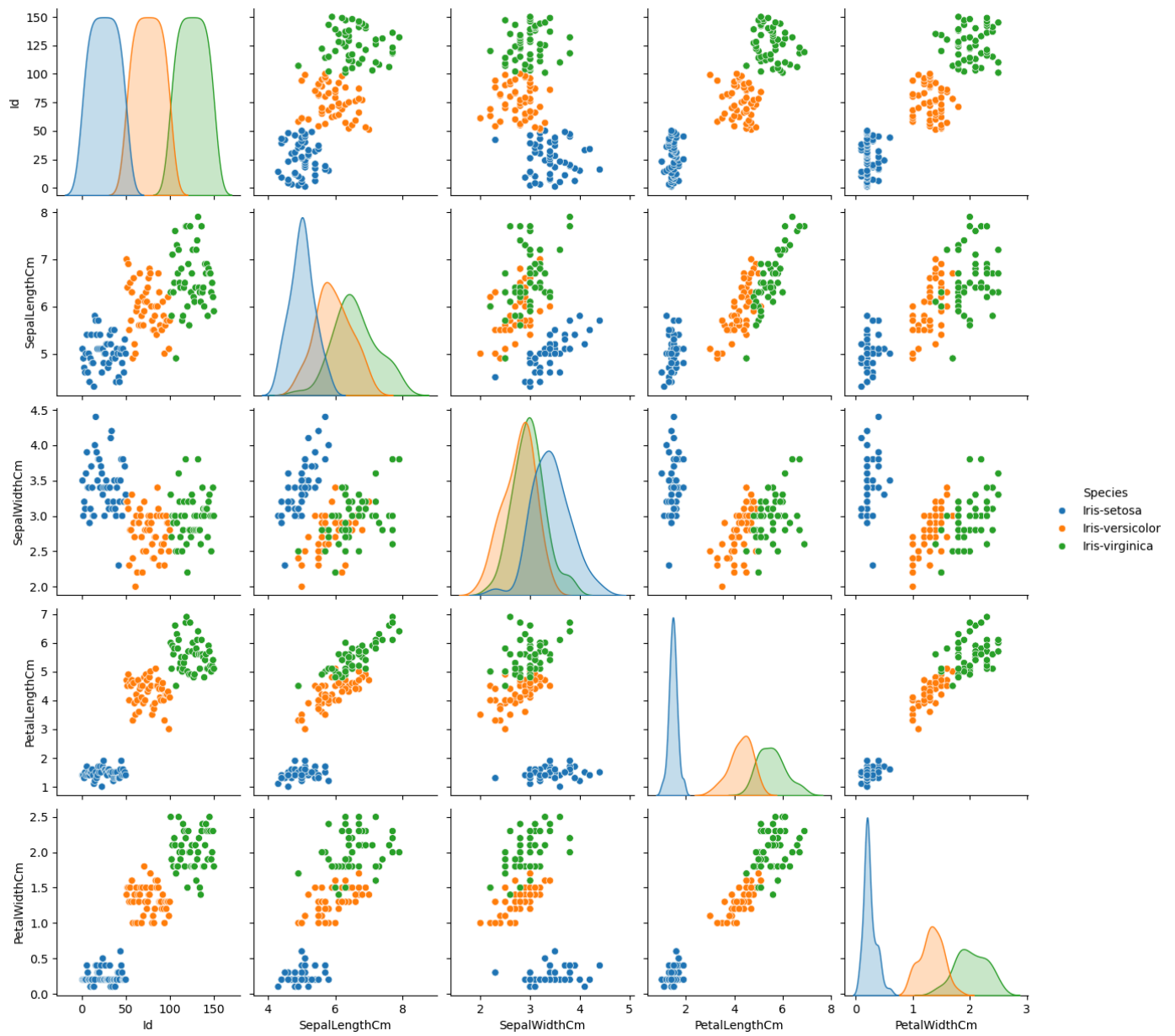


```
In [20]: data.drop('Species', axis=1).corr()
```

```
Out[20]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
Id	1.000000	0.716676	-0.397729	0.882747	0.899759
SepalLengthCm	0.716676	1.000000	-0.109369	0.871754	0.817954
SepalWidthCm	-0.397729	-0.109369	1.000000	-0.420516	-0.356544
PetalLengthCm	0.882747	0.871754	-0.420516	1.000000	0.962757
PetalWidthCm	0.899759	0.817954	-0.356544	0.962757	1.000000

```
In [21]: sns.pairplot(data, hue="Species")
plt.show()
```



In []: